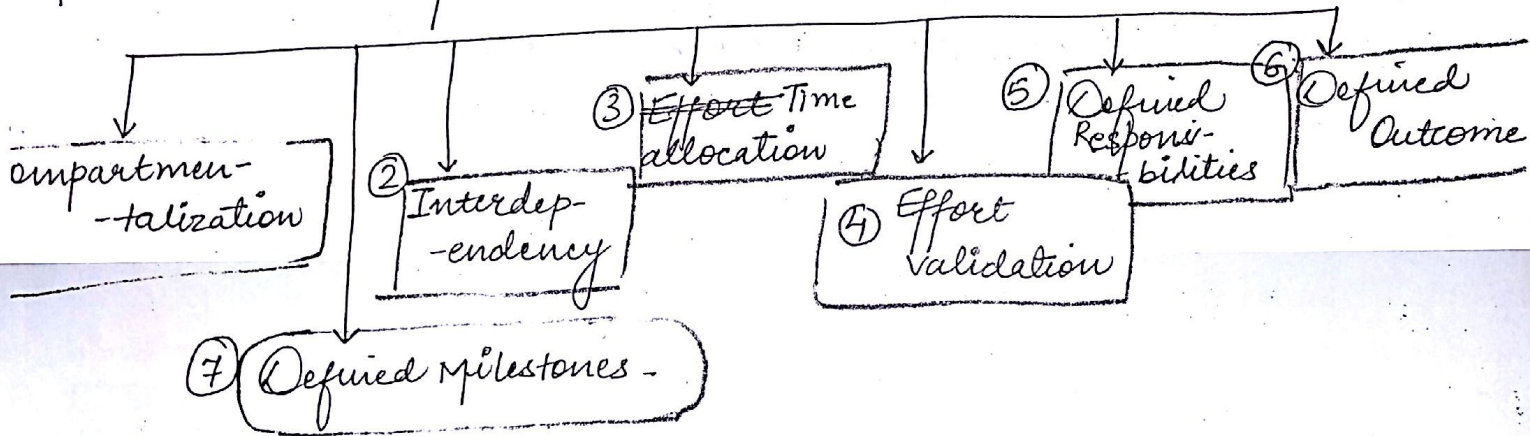


Project Scheduling & Tracking Techniques

Why projects are delayed?

- Unrealistic deadlines established out the team.
- Changing Customer requirements.
- Risks that were not Considered
- Miscommunication among project staff
- Failure by project Management to recognize project falling behind schedule.
- Underestimating the resources.

Software Project Scheduling Principles:-



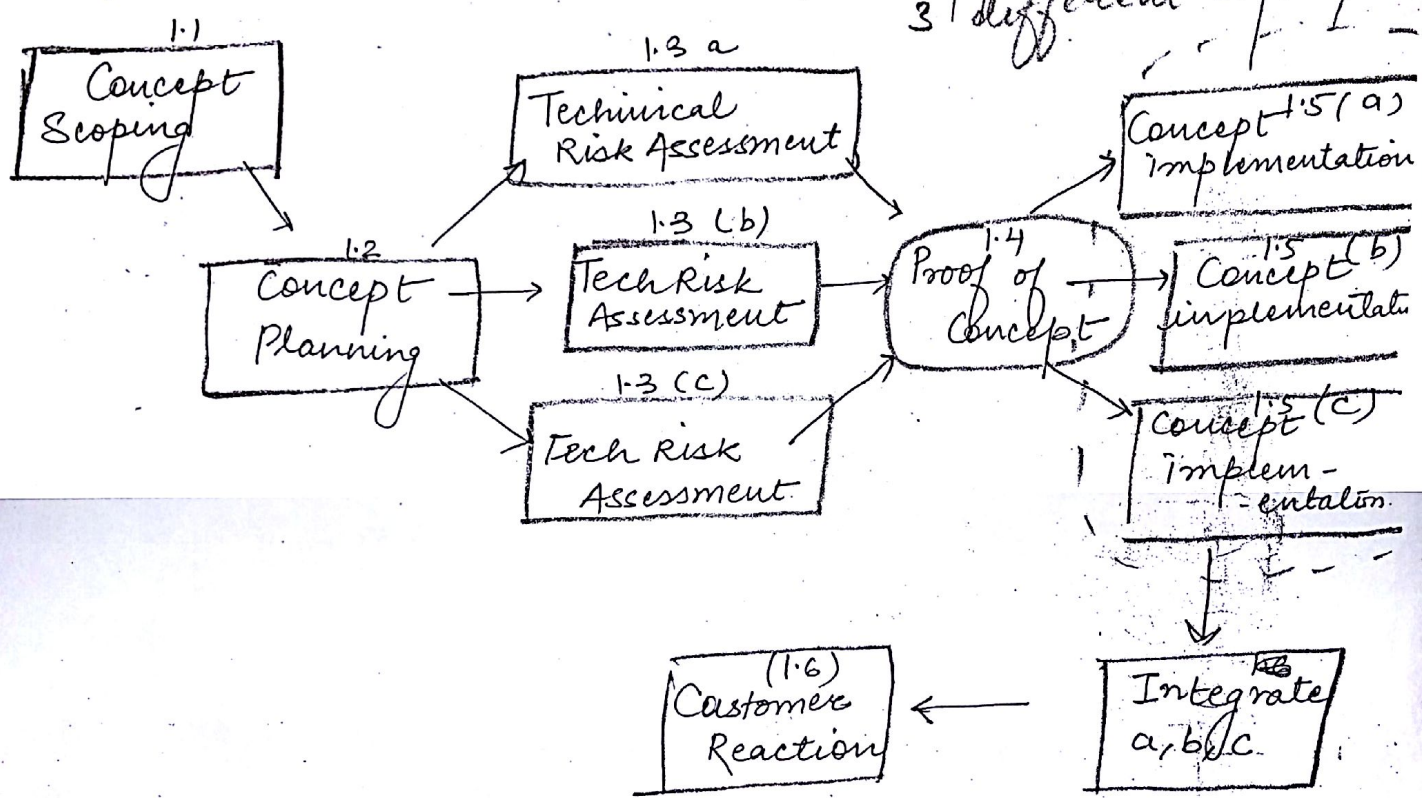
- 1) COMPARTMENTALIZATION :- product & process must be decomposed into no. of activities.
- 2) INTERDEPENDENCY :- tasks that can be completed in parallel must be separated from those must be completely serially.
- 3) TIME ALLOCATION :- Assign task starting & completion data.
- 4) EFFORT VALIDATION :- tasks distributed to the staff members to complete the tasks within the time estimated.
- 5) DEFINED RESPONSIBILITIES :- assign particular task to a specific team member.

DEFINED MILESTONE :- A milestone is accomplished to pass a quality review.

Software Effort Estimation

TASK NETWORK :-

1.5 tasks are applied in parallel 3 different concept



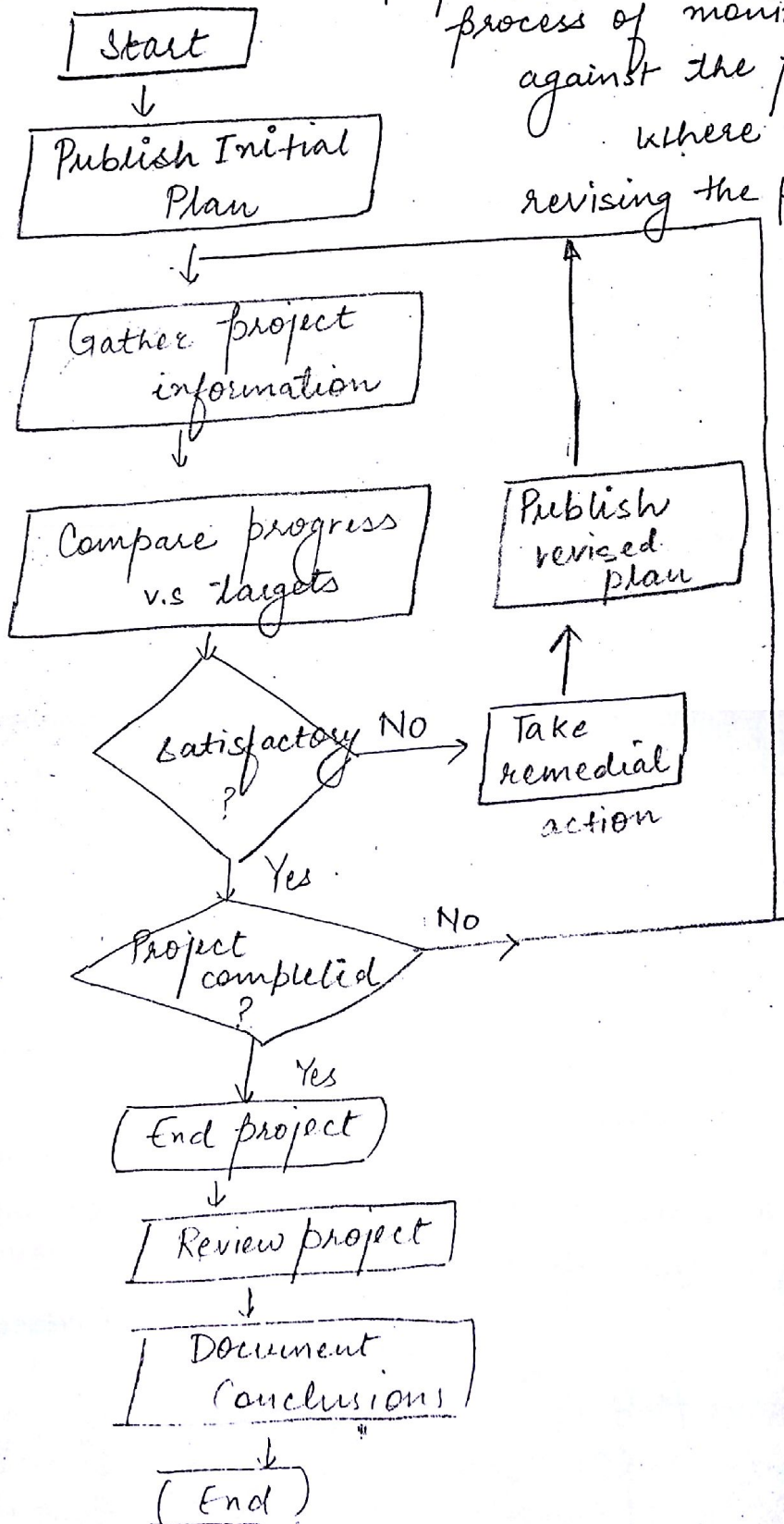
Monitoring & Controlling :-

This group consist of those processes performed to observe project execution so that potential problem can be identified in a timely manner and corrective actions can be taken, when necessary, to control the execution of the project.

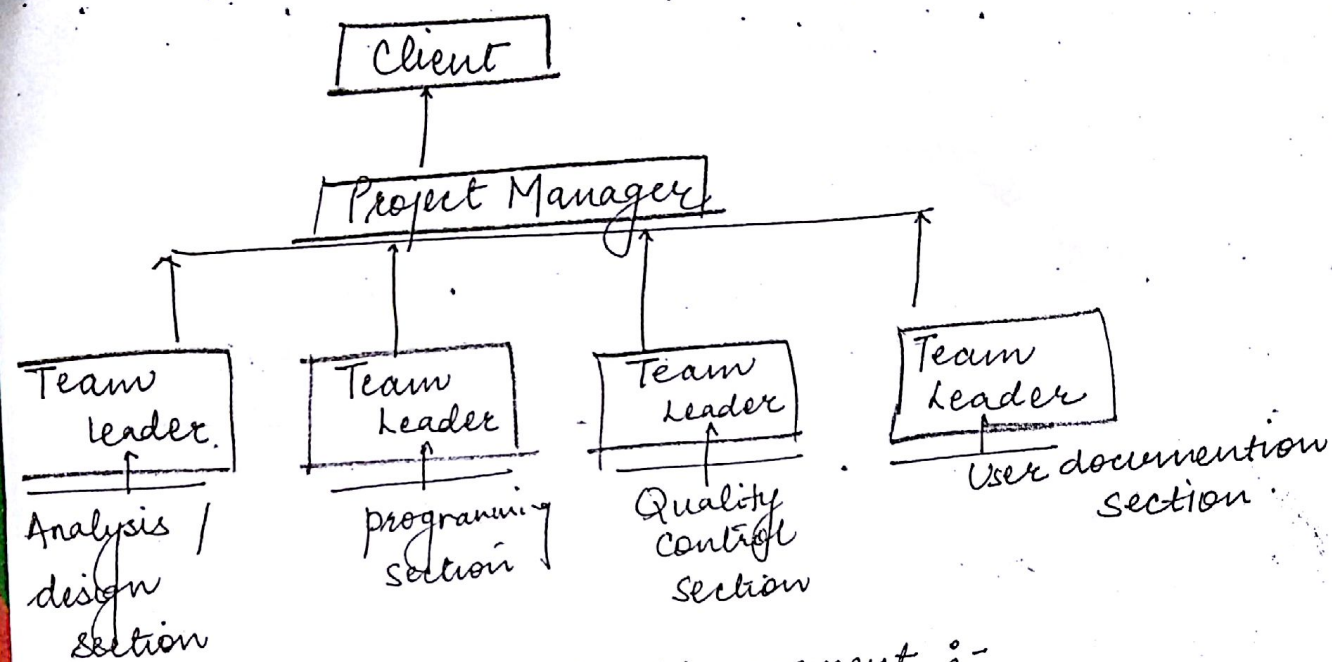
Monitoring and Control :-

Project Control Cycle

Shows how once the initial project plan have been published, project control is the continual process of monitoring progress against the plan, where necessary, revising the plan to take account.



Project Reporting Structures :-



K. Software Configuration Management :-

SCM is concerned with tracking & controlling changes to the S/W.

Why SCM or Use :-

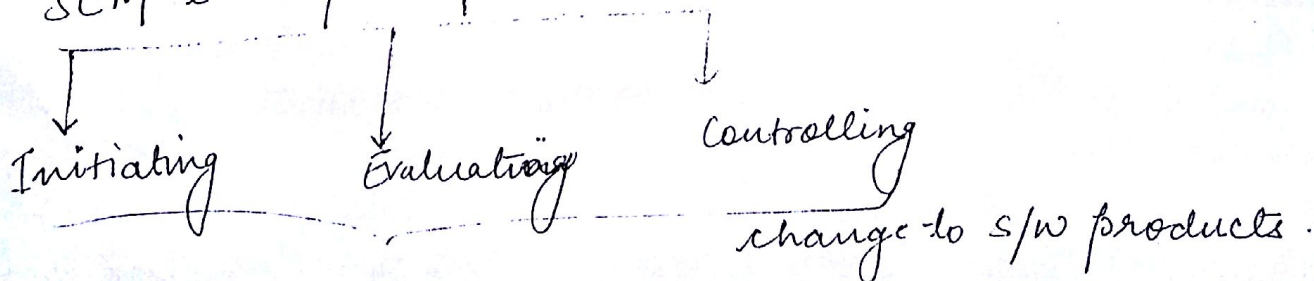
<i>The problem:-

- (i) Multiple people have to work on software that is changing.
- (ii) More than one version of the S/W has to be supported.

- > Need of Coordination
- > Manages evolving S/W systems.

Definition:- A set of management disciplines within the S/W engineering process to develop a Baseline.

SCM encompasses the discipline & techniques of



Purpose of SCM

(b)

Problem Associated With Concurrent Access :-

Important reason for Configuration management is to control the access to the different deliverables objects.

- 1) Undoing Change :- It becomes easy to undo some part of a revision to a certain version.
- 2) System Accounting :- It denotes keeping track of who made a particular change to a Configuration item, what change exactly made & when. Knowing → who, when & what.
- 3) Handling Variants :- Keep track of all variants and their versions and revision.

Accurate Determination of Project Status :- Configuration management Tool helps to track various deliverable objects so that the project manager quickly determine the current state of the project.

- 4) Preventing Unauthorized Access to the Work :- SCM helps to implement a controlled change process. It therefore becomes possible to prevent unauthorized change to the work.

Terminologies in SCM :-

Configuration :- Configuration of s/w is the state of various work products that are

under configuration control.

Work products under configuration control are usually referred to as the configuration items.

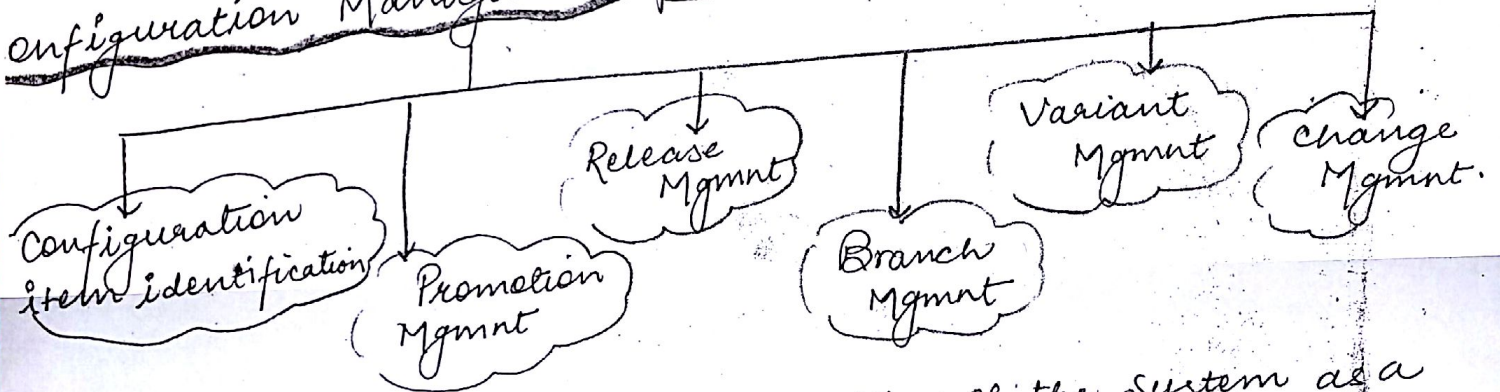
Version :- It is a numbering scheme that helps us to identify a specific configuration at a certain point in time.

Revision :- It is a numbering scheme that is used to identify the state of configuration at any time.

Baseline :- A baseline is a software configuration that has been formally reviewed & agreed upon & serves as a basis for further development.

Variants :- Variants are the versions that intended to co-exist.

Configuration Management Activities :-



Configuration Item Identification :- modelling of the system as a set of evolving components.

Promotion Mgmt :- is the creation of versions for other developers

3. Release Mgmt :- is the creation of versions for the clients & users.

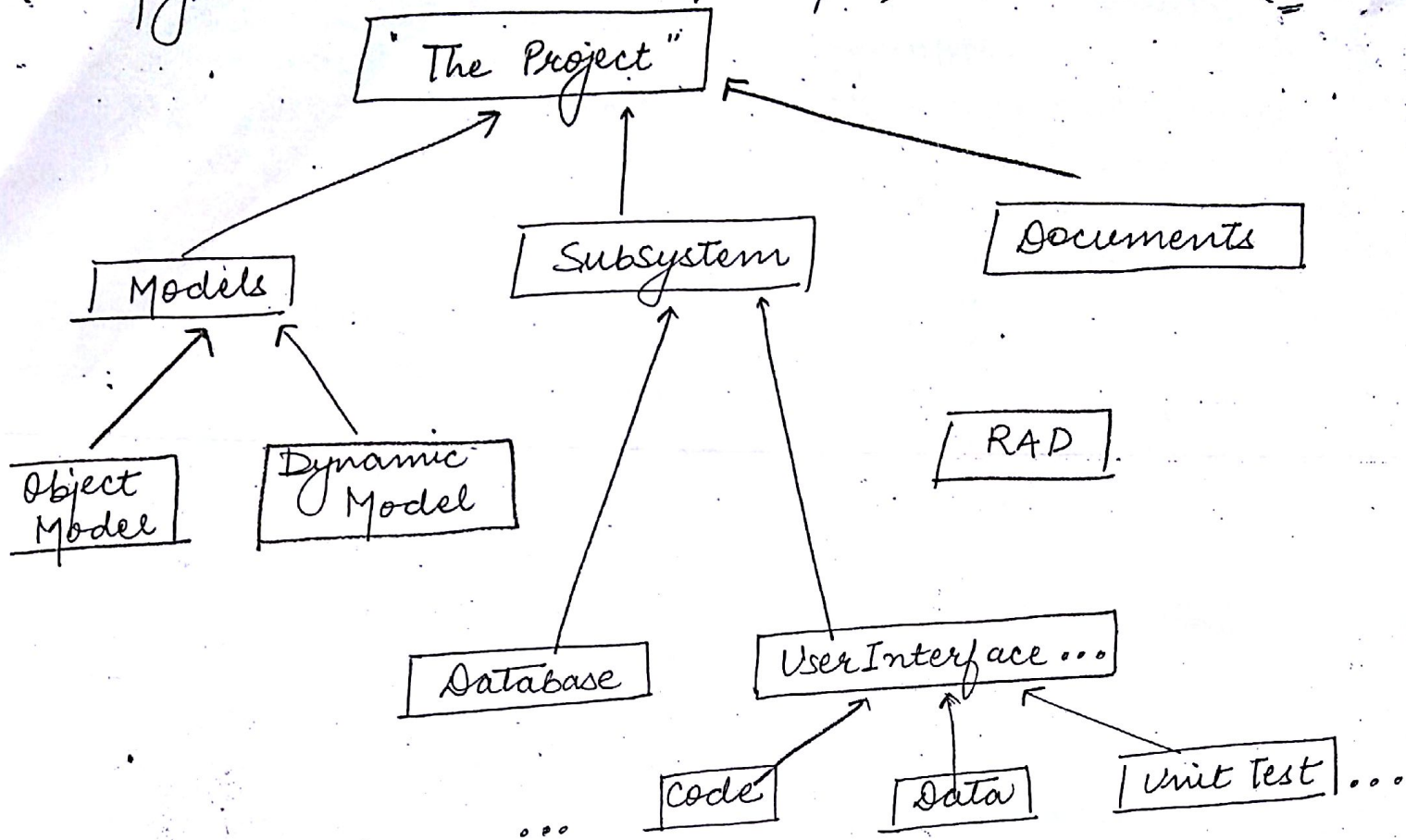
1. Change Mgmt :- is the handling, approval & tracking of change request.

5. Branch Mgmt :- is the management of concurrent development

Variant Mgmt :- is the management of versions intended to co-exist.

Configuration Item Tree (Example)

(c)

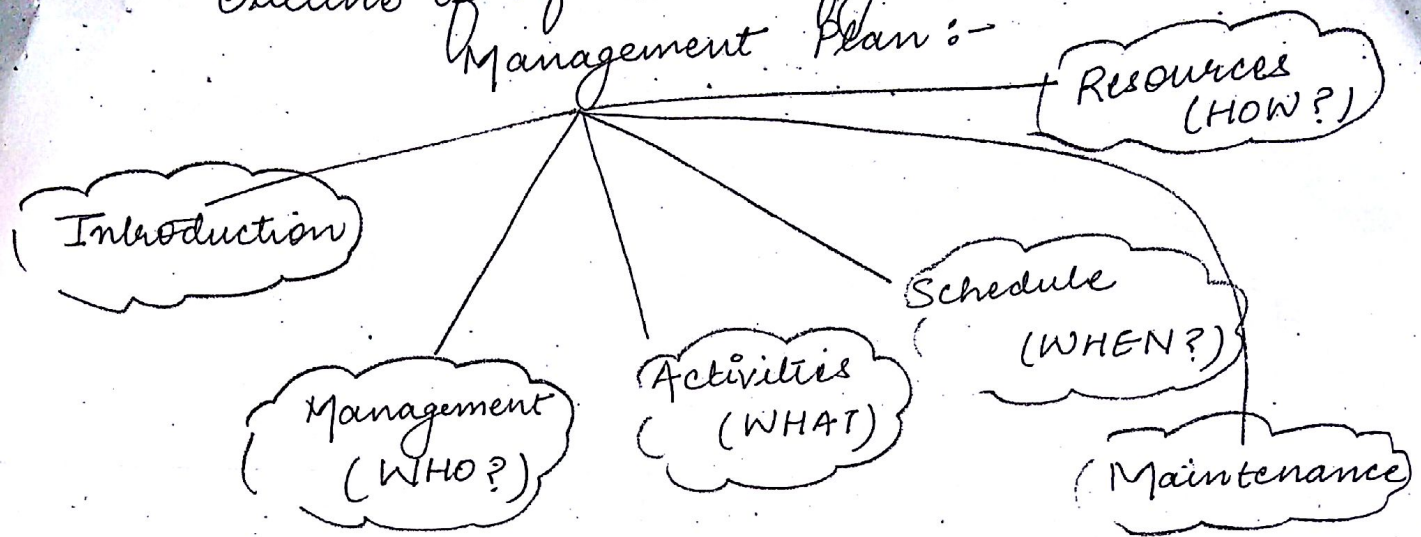


A. Tasks for Configuration Managers :-

- Define Configuration Items
- Define promote / release policies

- SCM Plan :-
- i) Defines the type of documents to be managed & a document naming scheme
 - ii) Define who takes responsibility for the CM procedure & creation of Baseline
 - iii) Define policies for ^{change} control & version management
 - iv) Describe the tools w/c should be used to assist CM process
 - v) Define the Configuration management database used to record configuration info.

Outline of Software Configuration Management Plan :-



1. Introduction :- Describes purpose, scope of application, key terms & references
2. Management (WHO?) Identifies the responsibilities & authorities for accomplishing the planned configuration mgmt activities
3. Activities (WHAT?) :- Identifies the activities to be performed in applying to the project.
4. Schedule (WHEN) :- Establishes the sequences & co-ordination of the SCM activities with project milestones.
5. Resources (HOW?) :- Identifies tools & techniques required for the implementation.
6. Maintenance :- maintain responsibility on how SCMP will be kept current during life cycle of the project.

* Monitoring & Control Process:-

2.

While project being executed, the project is being monitored, controlled by implementing level of insight corrective action

Project $\xrightarrow[\text{to ensure}]{\text{Measured}}$ that the project is within acceptable variance of $\begin{cases} \text{Cost} \\ \text{Schedule} \\ \text{Scope} \end{cases}$

Purpose:- to proactive in finding issues ahead of time & taking corrective action

Monitoring & Control Process:- process collects, measures, and disseminates performance info. & assesses measures to forecast potential items

1. Monitoring & Controlling Project Work :- includes the monitoring project risks, and taking controlling methods.

2. Integrated Change Control :- project corrective actions taken throughout the project

3. Scope verification :- ensures that project deliverables are accepted

4. Scope Control :- project scope is controlled

5. Schedule Control :- monitors the project schedule

6. Cost Control :- controls costs & changes to project budget

7. Performing Quality Control

8. Managing the project team :- tracks team members performance, provides feedback, resolves issues

9. Performance Reporting

10. Managing Stakeholders :- manage stakeholder opinion

to determine whether project is meeting quality standard

This process collects & distributes performance info.



Scheduling Activities :-

GANTT Chart

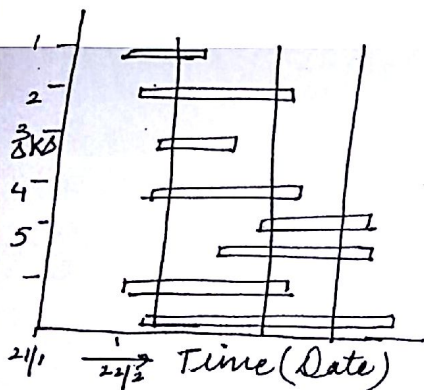
Schedule N/W Analysis

Critical Path Method :-

GANTT CHART :- Horizontal bar Chart plotted over time (e.g. days, weeks). Each activity is shown as bar.

↳ Depending on task dependencies & resource availability, these bars may be Sequential or parallel.

↳ Each Bar is plotted to start at the earlier possible start date



2. Schedule N/W Analysis :- Graphical display from left to Right across a page of all interrelationship b/w elements of work.

Initial Planning → Project Closure.

3. Critical Path Method :-

It is the Sequential string of activities that takes longest time to complete, recognize any dependencies b/w tasks in this sequences.

CPM is determined by adding the times of all activities in the critical path, Earliest time that the project can be completed.

* Prioritizing Monitoring:-

list of priorities applied in deciding levels of monitoring.

→ Critical Path Activities :- Any delay in any activity on critical path in the completion date for the project.

→ Activities with no free float :- a delay in activity with no free float.

→ Activities with less than a specific float :- If any activity has very little delay it might use up this float before the regular activity monitoring bring the problem to project manager.

High Risk activities :- A set high risk should be identified as a part of high risk.

Activities Using Critical Resources :- Activities can be critical because they are very expensive staff or other resources might be available for limited period. In an event, an activity that demands a critical resource required a high level of monitoring.

* Getting ^{project} Back to the Target:-

Every project at one or other time, subjects to delays and unexpected events.

One of the tasks of project manager is to recognize when it is happening and minimize the delay to the project team.

If delay occur there are two main strategies to consider when drawing up plan to bring a project back on target.

Strategies are

Shortening
the
critical path.

Altering the
activity precedence
requirements.

Change Control :-

It is a system approach to manage all changes made to a product or system.

Purpose :- is to ensure that no unnecessary changes are made, that all changes are documented, that services are not unnecessarily disrupted & all resources are used efficiently.

Steps for software Change Control

1) Documenting the change request :- when the client requests the change, that request is categorised & recorded, along with the informal assessments of importance of that change & the difficulty of implementing it.

Formal Assessment :- If the change request is accepted, a development team will be assigned. If the change request is rejected, the fact is documented & communicated to the client.

2) Planning :- Team responsible for the change creates a detailed plan for its design & implementation.

3) Designing & Testing :- Team designs the program for the s/w changes & tests.

Implementation & Review :- The team implements the program & stakeholders review the change.

Final Assessment :- If client satisfied → change was implemented satisfactorily & request closed.

If client is not satisfied → project is reassessed & step may be repeated.

- 4> Recommended corrective actions
- 5> Activity list and attribute updates
- 6> Updates to Project Management plan.

6> Cost control :-

- Input :-
- (i) Cost estimate updates
 - (ii) Cost Baseline updates
 - (iii) Performance measurements
 - (iv) Requested changes
 - (v) Recommended corrective actions
 - (vi) Updates to organizational process assets.

7> Performing Quality Control :-

- i) Quality Control measurement
- ii) Validate defect repair
- iii) Updates to the quality Baseline

> Managing the project team :-

- i) Requested changes
- ii) Recommended corrective & preventive actions

> Performance Reporting :-

- i> Performance Reports
- i> Requested changes
- ii> Recommended corrective actions

> Managing stakeholders :-

- (i) Resolved issues
- ii) Approved change requests
- iii) Approved corrective actions
- iv) Updates to project management plan

1. Monitoring & Controlling project work:-

Output include :-

1. Recommended corrective actions
2. Recommended preventive actions
3. Recommended defect repair
4. Request changes.

2. Integrated change Control:-

Input include :-

1. Approved change requests
2. Rejected change request.
3. Updates the project management plan.
4. Approved corrective and preventive actions
5. Approved defect repair
6. Deliverables.

3. Scope verification:-

Input include :-

- i) Accepted Deliverables
- (ii) Recommended corrective actions

4. Scope control:-

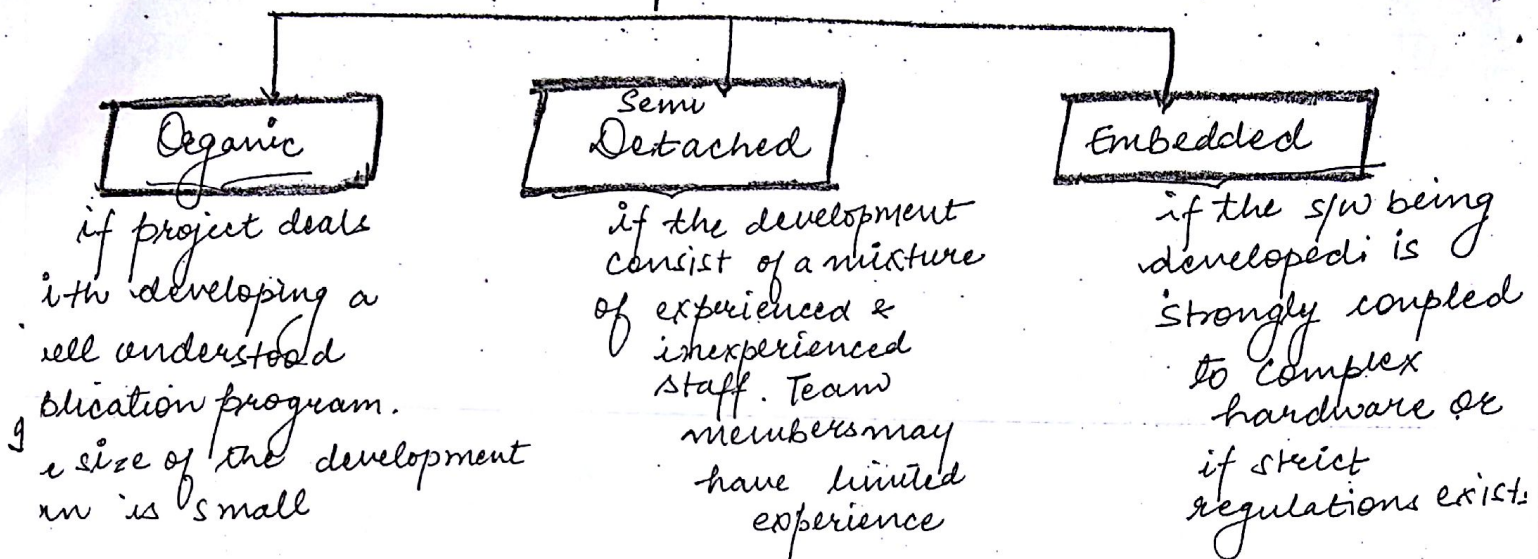
Input include :-

1. Updates to the project scope statement & Scope Baseline.
2. Update to Work Breakdown structure
3. Requested changes
4. Updates to organizational change assets.
5. Updates to project Management plan.

5. Schedule Control:-

- Output :-
- i> Updates to schedule model data
 - <ii> Performance measurement
 - <iii> Requested changes

Classification of SW development project.



COCOMO Model :-

Software ~~Software~~ Cost Estimation should be done through three stages :-

- Basic Cocomo Model
- Intermediate "
- Complete "

Basic COCOMO Model :-

It gives an approximate estimate of the project parameters.

$$\text{Effort} = a_1 \times (\text{KLOC})^{a_2} \text{ PM}$$

$$T_{\text{dev}} = b_1 \times (\text{Effort})^{b_2} \text{ Months}$$

KLOC = estimated size of the sw product, in Kilo lines of codes expressed

a_1, a_2, b_1, b_2 are constants for each category of sw products

T_{dev} is the estimated time of develop the sw

Effort is the total effort required to develop sw product

Effort estimation is expressed in units of person/month

(PM)

Good for :- Quick, early, rough order of magnitude estimates of sw costs.

Estimation of Effort

Organic : Effort = $2.4 \times (KLOC)^{1.05} PM$

Semi-Detached : Effort = $3.0 (KLOC)^{1.12} PM$

Embedded : Effort = $3.6 (KLOC)^{1.20} PM$

Estimation of Development Time

Organic = $T_{dev} = 2.5 (Effort)^{0.38} Months$

Semi-Detached = $T_{dev} = 2.5 (Effort)^{0.35} Months$

Embedded = $T_{dev} = 2.5 (Effort)^{0.32} Months$

Assume that the size of an organic type s/w product has been estimated to be 32,000 lines of code. Assume that the average salary of s/w eng. be Rs 15000/- per month. Determine the effort required to develop the s/w product and development time.

Ans: From Basic COCOMO Estimation

$$Effort = 2.4 \times (32)^{1.05} = 91 PM$$

$$Nominal Development Time = 2.5 \times (91)^{0.38} = 14 months$$

$$Cost \text{ required to develop the product} = 14 \times 15,000 = Rs 210,000/-$$

INTERMEDIATE COCOMO MODEL :-

Cost drivers are considered.

Project type	a	b
Organic	3.2	1.05
Semi-Detached	3.0	1.12
Embedded	2.8	1.20

Q. Explain if a s/w product is organic & it is estimated to be 8,000 LOC, the initial effort is calculated as:

$$E_i = 3.2 \times 8^{1.05}$$

$$= 28 \text{ person-months}$$

Ex-2: if a software product is considered embedded & is estimated to be 10,000 LOC, its nominal effort is as follows

$$E_i = 2.8 \times 10^{1.20}$$

$$= 44 \text{ person-months}$$

Duration Estimation :-

Project Type	a	b
Organic	2.5	0.38
Semi-Detached	2.5	0.35
Embedded	2.5	0.32

Example

Suppose the overall size of an organic S/W is estimated to 20,000

$$\text{Effort } E = 3.2 \times 20^{1.05} = 74 \text{ person-months}$$

$$\text{Duration } D = 2.5 \times E^{0.38} = 13 \text{ months}$$

If the monthly salary of each S/W developer is \$ 6,000/-

$$\boxed{\text{Costs } C} = 74 \times 6,000 = 440,000$$

$$\text{Average Staff size} = \frac{\text{Effort}}{\text{TDev.}}$$

$$\text{Productivity} = \frac{\text{Size}}{E} = \frac{\text{KLOC}}{\text{staff-month}}$$

Detailed COCOMO :- an extension of the intermediate model that adds effort multipliers for each phase of the project to determine the cost drivers impact on each step.

A major shortcoming of both the basic & intermediate COCOMO is :- they consider a S/W product as a single homogeneous entity. However most large systems are made up of smaller sub-systems.

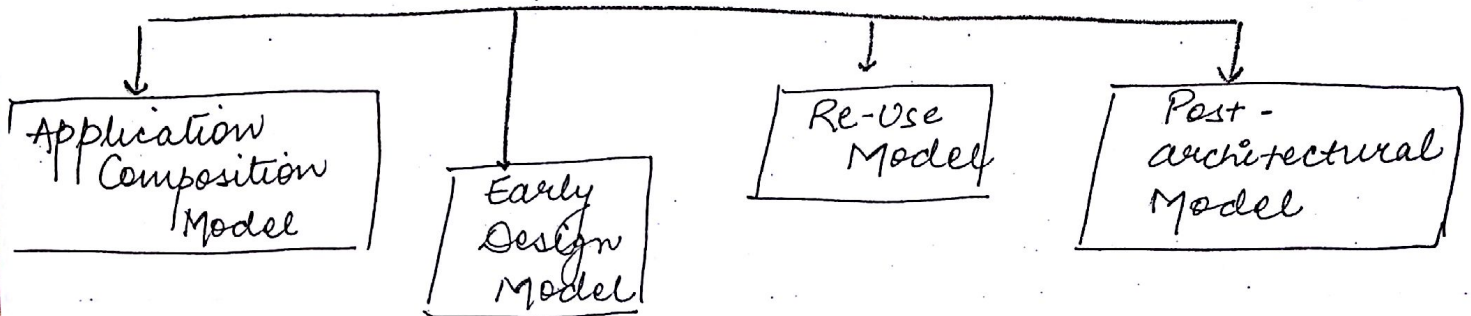
The Complete COCOMO model considers these difference in characteristics of the subsystem & estimates the effort and development time as the sum of the estimates for the individual subsystems.

COCOMO II Model :-

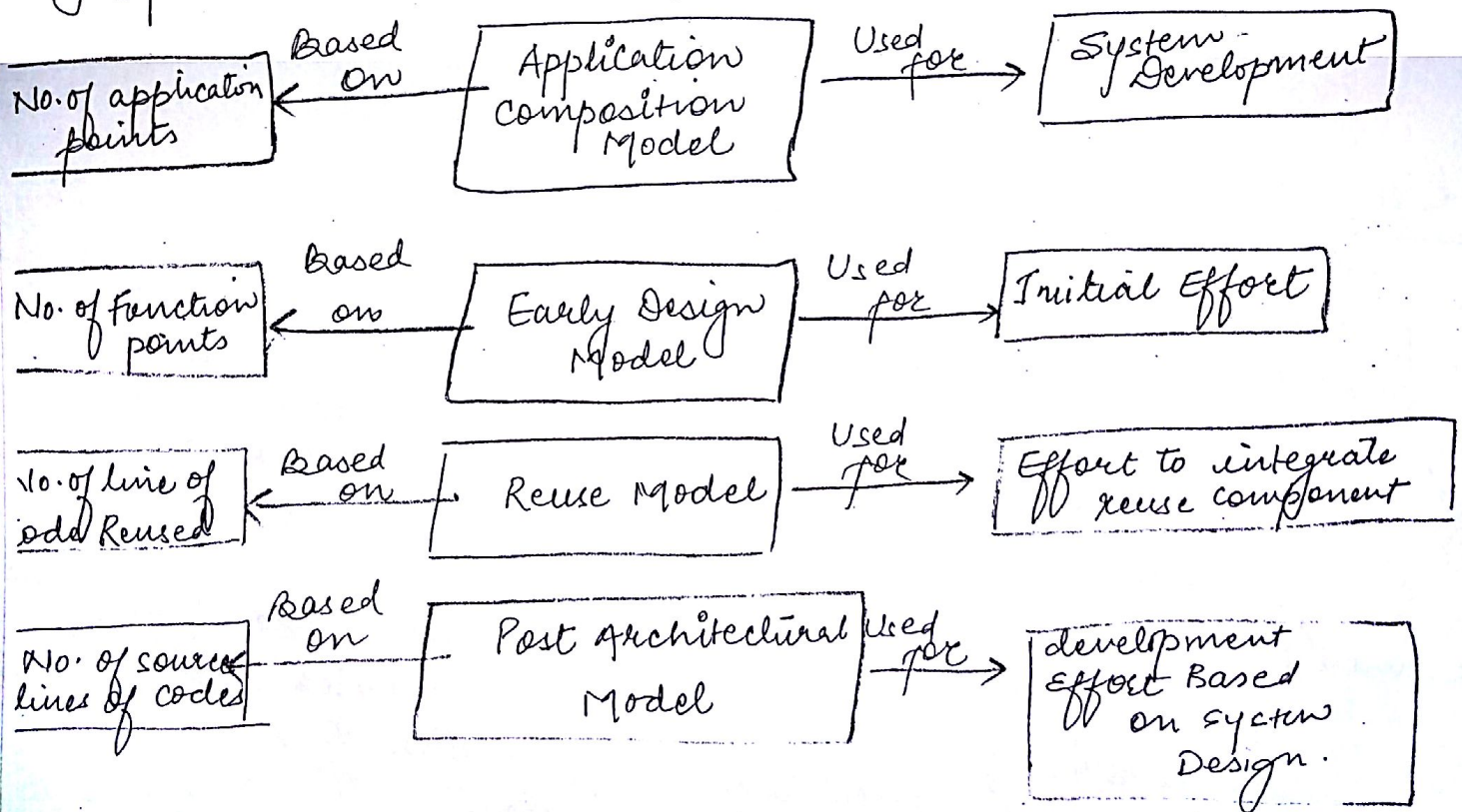
The Original COCOMO Model has been very successful but it doesn't apply to newer s/w development practices.

COCOMO II Targets modern s/w projects and will continue to evolve over next few years.

Its has 4 different model :-



Diagrammatic Representation :-



Application Composition Model :- for the earliest phase of spiral cycles. 3.

Ref Design Model :- for next phase of spiral cycles. Involves exploration of architectural (alternative or incremental development strategies.

Reuse Model :- for calculating reuse cost that indicates the amount of modification of codes reused.

Cost Architectural Model :- once the project is ready to develop it should have a life cycle architecture, which provide accurate info on cost drivers input & enables more accurate cost estimates.

Strategy Used in COCOMO II Model :-

- i> Preserve the openness of original COCOMO.
- i> Key the structure of COCOMO II to the future s/w marketplace sectors
- > Key the input/output of COCOMO II submodels to the level of information available.

Configuration Management Roles :-

- * Configuration Manager :- Responsible for identifying configuration items. CM can also be responsible for defining the procedures for creating promotions & releases.
- * Change control Board Member :- Responsible for approving or rejecting change requests.
- * Developer :- Creates promotions triggered by change request or the normal activities of development. The developer checks in changes & resolves conflicts.
- * Auditor :- Responsible for the selection & evaluation of promotions for release & for ensuring the consistency & completeness of this release.